

**Generic Feedback to  
January 2023/24  
2MVA Multivariable & Vector Analysis 06 25667 / 06 35172**

Overall, there were many good answers to the challenging exam!

**Q1a** involves calculating the rates and directions of steepest ascent, decent and directional derivative of a function at a given point. Most students did well, using the correct method. Common mistakes are: (i) not specifying the directions of the steepest ascent and the steepest decent, (ii) calculation errors, and (iii) not using the unit directional vector in calculating the directional derivative.

**Q1b** involves finding the tangent plane to a given surface at a specified point. Most students successfully obtained the correct answers for this part.

**Q1c** requires finding and characterizing the stationary points of functions. While most students were familiar with the leading minor test, some students did not obtain the stationary points correctly or failed to find all of them..

**Q1d** is to calculate the volume bounded by 6 surfaces. The solid is a part of the cylinder parallel to the  $z$ -axis, cut below by the  $oxy$  plane and above by the surface  $z = y^2$ . The solid is below the surface  $z = y^2$  and above the domain  $D$  in the  $oxy$  plane, which is bounded by  $x = y^2$  and  $y = x^2$ . With these considerations, many students obtained correct results using either a double integral or a triple integral.

**Q3a** involves integrating over a complex three-dimensional domain, which requires spatial visualization and the application of different coordinate systems. The question asks for the solution using (i) Cartesian coordinates, (ii) cylindrical coordinates, and (iii) spherical coordinates.

**Q3b** involves calculating the triple integral with the limits obtained in Q3a. Most students obtained correct answers, provided that the integral limits they found were accurate.

**Q3c** involves calculating triple integrals for even and odd functions over symmetrical domains. While many students obtained correct results, relatively fewer students provided proper justifications for their calculations. As the domain  $S$  is symmetric to the plane  $z = 0$  and the function  $xy^2 \sinh z$  is odd in terms of  $z$ , its integral over  $S$  vanishes. As the domain  $S$  is symmetric to the plane  $z = 0$ , its upper half is the domain  $T$  and the function  $xy$  is even in terms of  $z$ , we have  
The common mistakes for Q3 are setting the integral limits wrongly, failing to attempt or finish the answer,  
and failing to justify symmetrical properties used.

**Where students typically did well: Q1.**

**General suggestions for improving the results:**

Be patient during calculations to avoid mistakes.

Take care in setting integral limits, as incorrect limits can lead to erroneous solutions from the outset. Perform geometric analyses and, if necessary, draw figures of integral domains to logically determine the correct limits, rather than relying solely on intuition.

Revisit example questions in the notes, complete exercises from problem sheets, and clarify any uncertainties with the lecturer. Remember, practice leads to proficiency!